



Full Length Article

High resolution imaging of pressurised water supply lines

Saber Nasraoui^{a,b,*}, Moez Louati^c, Mohamed Salah Ghidaoui^c^a Department of Physics, Faculty of Sciences of Tunis, University Tunis El Manar 1060 Tunis, Tunisia, Laboratory of Mathematical Modeling and Multi-scale Simulation for Physics and Engineering, Tunisia^b Department of Water Supply, Sanitation and Environmental Engineering, IHE Delft, Netherlands^c Department of Civil and Environmental Engineering, Hong Kong University of Science and Technology, Clear Water Bay Campus, Hong Kong, China

ARTICLE INFO

Keywords:

Acoustic imaging
Time reversal
High-frequency acoustic waves
Pipeline
Water supply

ABSTRACT

This paper proposes a computationally efficient, nondestructive, and medium range guided wave imaging of pressurized water supply lines based on the time reversal (TR) technique. A send-receive transducer is inserted in the water column from a single access point and used to measure multi-input–multi-output (MIMO) pressure wave signals that propagate along the waveguide. The resulting high frequency (10 kHz–100 kHz) signals are processed by the TR-Multiple Signal Classification (TR-MUSIC) algorithm to provide high-resolution images the water pipe system. The resulting images reveal the pipe wall inner and outer condition with millimeter resolution. The proposed technique is tested and validated in lab as well as in field scale facilities on pressurized water-filled viscoelastic high-density polyethylene (HDPE) pipes. In particular, we successfully imaged (i) a straight 6.5 m long, water-filled 90 mm HDPE lab pipe containing a small blockage with thickness 7.65 mm and a length 97 mm and (ii) a 36.5 m long section water-filled 160 mm HDPE pipe containing three T-connections and a small blockage with thickness 16 mm and length 95 mm.

1. Introduction

Urban Urban Water Supplies Systems (UWSS)—a complex network of pipes that lie beneath our streets—are vital for life and economic productivity. Such systems often experience the formation of defects during their lifetime due to physical and/or chemical processes, including corrosion, failure of pipe lining, and deposition of suspended fine particles. Such defects often begin as a cluster of small defects on the inside of a pipe wall which we refer to hereafter as incipient defects. The growth of such incipient defects with time can eventually lead to bursts, leaks, and sizeable blockages, resulting in unacceptably large system water losses and excessive energy use. Recent studies estimate that 126 billion cubic meters are unaccounted for annually by water utilities around the world, for an equivalent monetary loss of USD 39 billion [1]. Until now, efforts to curb these massive water losses and energy wastage in existing UWSS has focused on non-intrusive and non-destructive technology development for detecting existing, fully-fledged, physical defects and operational inefficiencies. However, pinpointing incipient defects at an early stage for remediation prevents their progression into significant failures, offering a proactive approach to UWSS management.

Traditional internal sensor-based imaging systems are used to monitor and report the condition of a pipeline's internal surface and pipe wall and enable detection of corrosion sites, cracks, wall thickness, and weld condition, as well as existing leaks. Smart Pipeline inspection gauges (Smart PIGs) were originally developed by the energy pipeline industry and have only recently been emulated by

* Corresponding author.

E-mail address: nasraoui1110@gmail.com (S. Nasraoui).

water pipe tools like Smart Ball, Sahara, CCTV, and acoustic sonar [2–4]. These tools may also have on-board optical, magnetic, or acoustic sensors which, like Smart PIGs, must be launched into a water pipe and be retrieved through a suitably sized port, sometimes using a tether cable. PIG-based sensors are not autonomous or semi-autonomous (even in the energy pipeline sector); rather, they are propelled by the pipe flow or tethered. This technology is suitable for use in conduits that present no obstructions to free passage of inline sensing tools (eg., oil & gas pipelines). On the other hand, UWSS are often intricate networks of pipes with numerous pipe-diameter changes, partial-bore inline valves and abrupt changes in direction that do not permit the use of inline Smart PIGs. Additionally, inline Smart PIGs currently function as a slow, in-pipe survey tool, typically moving at a speed of approximately 0.3 m/s. Moreover, inline Smart PIGs involve significant costs and labor, require traffic diversion [2].

Less costly, non-disruptive, and more efficient techniques such as transient (acoustic)-based defect detection methods (TBBDM) have been intensively studied and developed [5–13]. Among these methods, the low frequency (<100 Hz) wave-based methods are unique for their fast speed (~ 1000 times faster than roving sensors) and broad coverage ($\sim$ kilometres), are both economical and applicable to all fully-developed faults. However, low frequency (LF) wave-based methods have poor location resolution (several meters) [6] and are limited to defects which are large enough to produce visible reflections since noise is strong at the LF band. Since the achievable resolution is directly proportional to the probing wavelength, high frequency (HF) wave-based methods have been developed to achieve a resolution in the order of centimeters [14,15]. Yet, these HF wave-based methods remain applicable only for fully fledged defects (i.e., through-wall defects such as leaks and also defects with a radial length scale of the order of 15 % of the pipe diameter or larger). These limitations underscore a significant gap: existing methods cannot reliably detect and image incipient defects or small structural anomalies at the millimeter scale. To address this gap, we propose a novel pipeline imaging technique based on the time reversal (TR) property of high-frequency acoustic waves (10–100 kHz) enabling subwavelength resolution.

This method would permit UWSS management to move from reactive to proactive mode, where anomalies with length scales of the order of millimetre are detected and imaged (not just localized). The proposed imaging technique serves as a decision support tool for timely maintenance and mitigation measures. The produced image can reveal the pipe wall inner and outer condition, incipient as well as existing defects at a scale up to millimeters and small (5 to 10 %) change in wall stiffness or mass. The proposed pipeline imaging technique is based on the wave time reversal property that has been experimentally proven in UWSS [16]. This technique specifically advances the transient-time-reversal by demonstrating its capability to image small-scale defects in pressurized, viscoelastic, water-filled HDPE pipes—an application not previously achieved. The technique operates through a forward step that actively generates HF waves, which are sensed, followed by a Time Reversal step where subspace noise is reversed and re-emitted into a UWSS model. This approach allows for detailed imaging of the pipe wall's inner and outer conditions, enabling proactive UWSS maintenance. The technique operates through a forward step where actively-generated high frequency waves in UWSS are sensed, followed by a Time Reversal step where the chronological order of the subspace noise is reversed and re-emitted into a model of the UWSS. The proposed technique is tested and validated in lab as well as in field scale facility comprised of pressurized water-filled viscoelastic HDPE pipes.

2. Fundamental principle and Time-Reversal imaging algorithm

Consider a water-filled pipe with a cross section of radius R as sketched in Fig. 1. We use the cylindrical coordinates, (x, r) , where r is defined along the radius in the pipe cross section and the cylindrical coordinate x is defined along the cylinder axis.

2.1. Multistatic response matrix

The MIMO data is obtained via an array of transducers in which the complete set of time-domain signals for every pair of transmitter–receiver elements in the array is collected. Fig. 1 shows the locations of M —transducers array placed along the pipe centerline

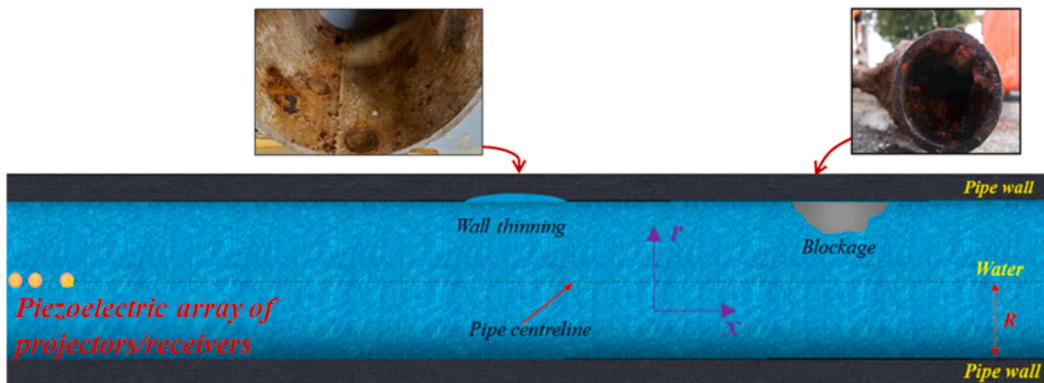


Fig. 1. Sketch of a water-filled pipe system containing multiple defects and a piezoelectric array. It shows the location of M —piezoelectric sensor array along the pipe centerline axis. The piezoelectric elements are used not only as transmitters but also as receivers. Each element in the piezoelectric array is excited in turn, and all the array elements receive response signals.

denoted as $\mathbf{x}_\ell = (x_\ell, r_\ell)$ where x_ℓ and r_ℓ represent respectively the axial and the radial coordinates of the ℓ^{th} location, and $\ell = 1, 2, \dots, M$. The data is collected as follows: at each of the ℓ^{th} transducer location, an acoustic source signal $S(t)$ is projected, and the system response is measured at all locations within the array at a sampling rate F_s (as the rule of thumb, the F_s is ten times the maximum injected frequency). The digitized time-domain signals $P(n/F_s, \mathbf{x}_i | \mathbf{x}_j)$ from each transmitter–receiver combination are grouped into a $(M \times M \times N)$ 3D matrix $\mathbf{K}_{3D}$ (Fig. 2) where $i = 1, 2, \dots, M, j = 1, 2, \dots, M, n = 1, 2, \dots, N$, and N is the signal length in time, $\mathbf{x}_j$ is the source location and $\mathbf{x}_i$ is the receiver location.

This paper focuses on imaging with high frequency (10kHz–100 kHz). The low level of noise in this frequency range [17] allows the use of the single-frequency versions of the TR-MUSIC algorithms and develop real time pipe imaging technique. In this approach, we perform the fast Fourier transforms (FFTs) on the entries in the 3D matrix $\mathbf{K}_{3D}$, which contains the time-domain signals collected by the hydrophone array along the pipe. This transformation converts the time-domain signals into the frequency domain, resulting in the frequency-domain 3D matrix $\hat{\mathbf{K}}_{3D}$. From $\hat{\mathbf{K}}_{3D}$, we then extract the 2D multistatic response matrix $\mathbf{K}(\omega)$, which represents the pipe system's response at each specific frequency ω . This frequency-domain matrix $\mathbf{K}(\omega)$ is subsequently used in the TR-MUSIC algorithm to generate a detailed, real-time image of the pipe system. For example, when using an acoustic source with a Gaussian-modulated sine pulse, the imaging technique is applied to $\mathbf{K}(\omega_0)$, which is the 2D multistatic response matrix of the pipe system at the central frequency ω_0 . Here, ω_0 represents the central frequency of the injected source, which corresponds to the peak of the frequency response signal.

2.2. Time-Reversal operation

In this paper the MUSIC approach for time-reversal imaging is considered. This approach was first proposed in [18] and named ‘time-reversal MUSIC’ because both time-reversal methods [19] and MUSIC can be explained using the eigenvalues of the time reversal operator [20]. The MUSIC algorithm, a high-resolution signal processing technique, works by decomposing the multistatic response matrix $\mathbf{K}(\omega)$ (which can be provided experimentally from measurement the data collected on the hydrophone array) into orthogonal signal and noise subspaces using singular value decomposition (SVD). This mathematical framework is particularly useful for time-reversal imaging as it exploits the inherent properties of the Green's function and the orthogonality of subspaces. The MUSIC algorithm, a high-resolution signal processing technique, works by decomposing the multistatic response matrix (which can be obtained experimentally from the data collected by the hydrophone array) into orthogonal signal and noise subspaces using singular value decomposition (SVD). This mathematical framework is particularly useful for time-reversal imaging, as it leverages the inherent properties of the Green's function and the orthogonality between the signal and noise subspaces. The signal subspace is constructed analytically using the Green's function, which models the propagation of waves through the medium. The Green's function formalism helps to represent the expected response of the system to signals originating from potential scatterer locations. Once the signal is analytically constructed, it is projected onto the noise subspace. Specifically, the imaging function relies on projecting test locations onto the noise subspace, which enables a precise identification of the spatial distribution of scatterers.

More details about the imaging algorithm steps are given in the following.

i) Singular Value Decomposition:

Consider a multistatic response matrix $\mathbf{K}(\omega)$, which can be provided experimentally from measurement the data collected on the hydrophone array. The singular value decomposition (SVD) of $\mathbf{K}(\omega_0)$ is:

$$\mathbf{K}(\omega_0) = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \quad (1)$$

where the superscript ‘ H ’ is the conjugate transpose, $\mathbf{U}$ and $\mathbf{V}$ are matrices whose columns are the singular vectors μ_i and $\mathbf{v}_i$, while $\mathbf{\Sigma} = \text{diag}\{\sigma_1, \sigma_2, \dots, \sigma_M\}$, is a diagonal matrix with singular values σ_i , arranged in order of decreasing magnitude. The singular vectors μ_i and $\mathbf{v}_i$ are both a mixture of signal and noise eigenvectors, and they are orthogonal to each other. In practice, because of the presence of experimental noise, additional singular values may become significant, which makes it difficult to distinguish the signal and noise subspaces. In these cases, a threshold criterion can be chosen to divide the signal subspace from the noise subspace [21]. The value of this threshold is empirical and so should be tuned depending on the system and defects of interest. Here, in the case of pipe system imaging a threshold of 10 % of the maximum singular value at each frequency was found to give accurate blockage image [21]. The

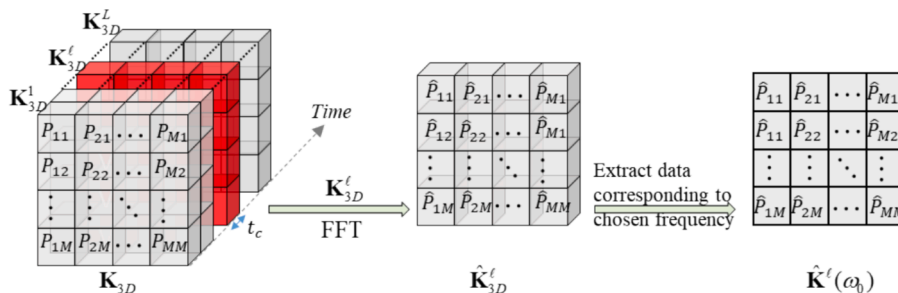


Fig. 2. Experimental transfer matrix arrangement. P_{ij} denote the transient response at time t and location $\mathbf{x}_i$ due to a wave source at $\mathbf{x}_j$.

sensitivity of the chosen threshold (10 %) is addressed, and the results provide that for a perturbation of a few percent, the output TR-MUSIC blockage image remains unchanged. Under the threshold criterion the signal and noise components are extracted as follows: $\mathbf{U} = [\mathbf{U}_{\text{signal}} | \mathbf{U}_{\text{noise}}] = [\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_m | \boldsymbol{\mu}_{m+1}, \dots, \boldsymbol{\mu}_M]$ and $\mathbf{V} = [\mathbf{V}_{\text{signal}} | \mathbf{V}_{\text{noise}}] = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m | \mathbf{v}_{m+1}, \dots, \mathbf{v}_M]$. where, $\mathbf{U}_{\text{signal}}$ and $\mathbf{V}_{\text{signal}}$ are the left and right signal subspaces, $\mathbf{U}_{\text{noise}}$ and $\mathbf{V}_{\text{noise}}$ are the left and right noise subspaces, and m is the number of columns of the signal subspaces matrix, which is threshold dependent.

ii) Imaging function:

A first-order Newmann series solution is considered to obtain the theoretical model for multi-scatterers in a pipe system where weak scattering strength is assumed [22,23]. In addition, reference [20] shows that the SVD decomposes the response matrix into a set of eigenvectors associated with both signal (Green's function-related) and noise subspaces. Consequently, the theoretical singular vector ($\boldsymbol{\mu}^{\text{Model}}$) from the signal subspace associated to an arbitrary scatterer at $\mathbf{x}_{\text{sc}} = (x_{\text{sc}}, r_{\text{sc}})$ can be written as:

$$\boldsymbol{\mu}^{\text{Model}}(\omega_0, \mathbf{x}_e | \mathbf{x}_{\text{sc}}) = [G(\omega_0, \mathbf{x}_1 | \mathbf{x}_{\text{sc}}), G(\omega_0, \mathbf{x}_2 | \mathbf{x}_{\text{sc}}), \dots, G(\omega_0, \mathbf{x}_M | \mathbf{x}_{\text{sc}})]^T \quad (2)$$

where G is the Green's function of the pipe system medium defined as [24,25]

$$G(\omega_0, \mathbf{x}_e | \mathbf{x}) = \frac{i}{2\pi} \sum_{\eta} \frac{J_0(k_{r\eta} r_e) J_0(k_{r\eta} r)}{R k_{x\eta} \left(1 - \frac{\omega_0^2}{R^2 k_{r\eta}^2 Z^2(\omega_0, k_{r\eta})} \right) J_0^2(R k_{r\eta})} \exp(-i k_{x\eta} |x_e - x|) \quad (3)$$

$G(\omega_0, \mathbf{x}_e | \mathbf{x})$ represents the system frequency response at $\mathbf{x}_e$ given a source at $\mathbf{x}$, for a given frequency ω_0 , and comprises the sum of multiple propagating modes η , each described by the modal axial and radial wavenumbers, $k_{x\eta}(\omega_0)$ and $k_{r\eta}(\omega_0)$, respectively. J_0 is the Bessel function of the first kind of order zero, and $Z(\omega_0, k_{r\eta})$ is the impedance of the pipe wall, which is evaluated using the shell theory model [14,26]. Given the orthogonality property between the signal and noise subspaces, the scatterers locations ($\mathbf{x}_{\text{sc}}$) are associated with the locations $\mathbf{x}$ at which the inner product between the green function vector $\mathbf{g}(\omega_0, \mathbf{x}_e | \mathbf{x}) = [G(\omega_0, \mathbf{x}_1 | \mathbf{x}), G(\omega_0, \mathbf{x}_2 | \mathbf{x}), \dots, G(\omega_0, \mathbf{x}_M | \mathbf{x})]^T$ and the noise subspace ($\boldsymbol{\mu}_i$ ($i = m + 1, \dots, M$)) is zero, where $\mathbf{x} = (x, r)$ scans the whole pipe domain space. Thus, the imaging function $\varphi(\mathbf{x}, \omega_0)$ can then be constructed as follows

$$\varphi(\mathbf{x}, \omega_0) = \left[\sum_{i=m+1}^M |\langle \boldsymbol{\mu}_i, \mathbf{g}(\omega_0, \mathbf{x}_e | \mathbf{x}) \rangle|^2 \right]^{-1} \quad (4)$$

where the strength of $\varphi(\mathbf{x}, \omega_0)$ at different $\mathbf{x}$ provide the pipe local image corresponding to the measured transfer matrix $\mathbf{K}(\omega_0)$. The set of $\mathbf{x}$ values where $\varphi(\mathbf{x}, \omega_0)$ is maximum provides the special region that defines the scatterers (i.e. defects in the pipe).

iii) Local Imaging function:

TR-MUSIC has a poor capability of detecting multiple scatterers, especially when the number of the sensors in the array is small [27]. In order to deal with the forgoing issue, an enhanced monitoring strategy based on the local TR-MUSIC algorithm [28] is applied. This means that the acoustic response from the pipe system is split into a set of sub-responses. Each sub-response presents the scattered signal from a local region in the pipe. In addition, since the scattered signals of different time periods are cut out for a local region processing from zero, this is equivalent to scanning the pipe domain from near to far according to the distance from the location of the receiver. The process of the local TR-MUSIC algorithm is similar to the short-time Fourier transform. First, a moving Hanning window (MHW) in time is applied to $\mathbf{K}_{3D}$ to obtain a set of 3-D local matrices $\mathbf{K}_{3D}^{\ell}$ where $\ell = 1, 2, \dots, I$, and $I = N/t_c$, with t_c is the time interval of the MHW (see Fig. 2) [28]. Second, each local matrix $\mathbf{K}_{3D}^{\ell}$ is transformed in the frequency domain using the Fast Fourier transform (FFT) and the result is $\hat{\mathbf{K}}_{3D}^{\ell}$. From $\hat{\mathbf{K}}_{3D}^{\ell}$, we then extract the local multistatic response matrix $\hat{\mathbf{K}}^{\ell}(\omega_0)$. Thus, there exists t_c -dependent singular value decomposition nonzero singular values. Herein, the t_c is chosen so that the number of scatterers (i.e., the number nonzero singular values) does not exceed the total number of sensing array elements.

Finally, the time reversal operation is applied to every frequency-transformed local matrix for a given frequency $\hat{\mathbf{K}}^{\ell}(\omega_0)$ (local multistatic response matrix), performing a set of imaging functions $\varphi^{\ell}(\mathbf{x}, \omega_0)$. The full pipe imaging function $\varphi(\mathbf{x}, \omega_0)$ is obtained by summing all local imaging functions $\varphi^{\ell}(\mathbf{x}, \omega_0)$ for $\ell = 1, 2, \dots, I$, which is written as

$$\varphi(\mathbf{x}, \omega_0) = \sum_{\ell} \varphi^{\ell}(\mathbf{x}, \omega_0) \quad (5)$$

The strength of $\varphi(\mathbf{x}, \omega_0)$ at different $\mathbf{x}$ provides the full pipe image.

3. Testing the proposed imaging method using synthetic data

3.1. Noise-free synthetic data: Rigid pipe case

This section provides an example to verify the effectiveness and feasibility of the developed TR-based imaging algorithm using synthetic data generated by the analytical model developed in Ref ([14]). The test case consists of an unbounded rigid pipe with radius $R = 0.0393$ m containing a small blockage located 2.36 m away from the sensors array (see Fig. 3). The blockage was simulated in the two-dimensional pipe domain by aligning closely 70 spaced scattering points. In this example the MIMO data is obtained via an array of 10 transceivers placed along the pipe centerline and spaced every 2 cm. The simulated blockage is drawn with small dots (Fig. 3b) and the array of 10 transceivers is drawn with small circles (Fig. 3a). The blockage is modeled as a discrete set of points. During the simulation process, it is numerically represented as a collection of scattering points to compute the reflection from the entire blockage. The reflection is simulated by calculating the interaction of the incident wave field with each scattering point using the Green's function formalism[14]. Initially, a uniform scattering strength of one was assigned to each point on the blockage. When noise is introduced, the signal-to-noise ratio is controlled by adjusting the amplitude of the noise.

The wave speed in water is assumed to be constant at 1480 m/s. The excited wave source signal $S(t)$ is a Gaussian-modulated sine wave with central frequency $f_0 = 55$ kHz (or $\omega_0 = 2\pi f_0$) and a narrow frequency bandwidth $\text{FBW} = [0.9f_0 \text{ to } 1.1f_0]$ (i.e. the minimum probing wavelength $\lambda \sim 2.4$ cm) [29,30]. The sampling rate is 1 MHz, and the length of the time signal of the system response is set to $T = 0.01$ s, which is enough to capture the reflection from the blockage. Fig. 4.a shows the scattered signal received by the 10th sensor (i.e., the one nearest to the blockage) when the source is at the first position of the transducers array.

In this test case the number of the sensor array (10 sensors) is smaller than the number of the scatterers (70 scattering points). Therefore, the local TR-MUSIC algorithm is applied. The time interval t_c of the Hanning window is taken to be $T/30$ in order to ensure that the number of scatterers is smaller than the number of transducer elements (i.e., 10). Moreover, one can choose t_c smaller than $T/30$, but this increases the time of the imaging computation without gain in terms of accuracy. In fact, a Hanning window with $t_c = T/40$, and $t_c = T/60$ has also been applied (not shown here for brevity), and the imaging accuracy is the same as the $T/30$ case.

For the time window interval $t_c = T/30 = (0.01/30)$ s, the scattered signal is divided into 30 sub-scattered signals in which there are nine signals with significant amplitudes (see Fig. 4b). From each received signal, nine sub-scattered signals caused by the blockage are subtracted, then the resulting nine transfer matrices are substituted into the TR-MUSIC algorithm, performing nine imaging functions $\varphi^\ell(\mathbf{x}, \omega_0)$, $\ell = 1, 2, \dots, 9$ plotted in Fig. 5. Combining Fig. (5.1)–(5.9) gives the accurate image of the blockage in the considered pipe system (see Fig. 6).

3.2. Effect of noise: Rigid pipe case

A zero-mean Gaussian white noise is added to the synthetic data of the previous test case. Therefore, the multistatic response matrix can be written in the form

$$\mathbf{K}' = \mathbf{K} + \mathbf{N} \quad (6)$$

where $\mathbf{K}$ is the noise-free multistatic response matrix and $\mathbf{N}$ is the added zero mean complex Gaussian noise matrix. The signal-to-noise ratio (SNR) in decibels is defined by

$$\text{SNR} = 20 \log_{10} \left(\frac{\|\mathbf{K}\|}{\|\mathbf{N}\|} \right) \quad (7)$$

where $\|\cdot\|$ is the usual L_2 matrix norm.

Fig. 7 shows the pipe image of the considered blockage, computed from a single realization of noisy data having a signal-to-noise ratio of roughly 22 dB. The effect of the noise is evident from the Fig. 7 in the form of spurious maxima surrounding the maxima due to

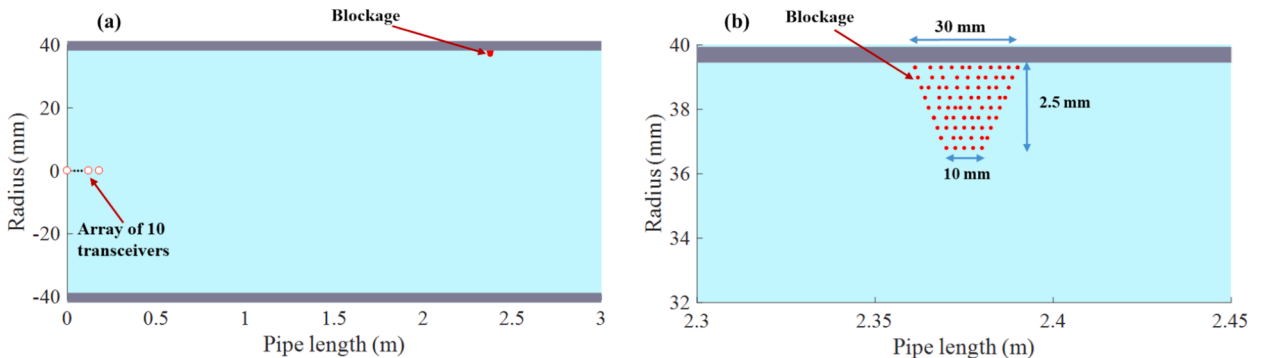


Fig. 3. (a): Sketch of the considered pipeline system for imaging. (b): Zoomed version.

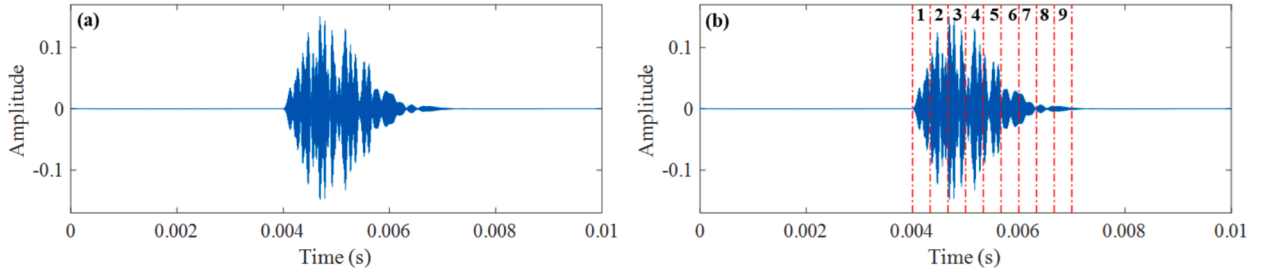


Fig. 4. Scattered signal received by the 10th sensor when the source is acting on the first transducers array.

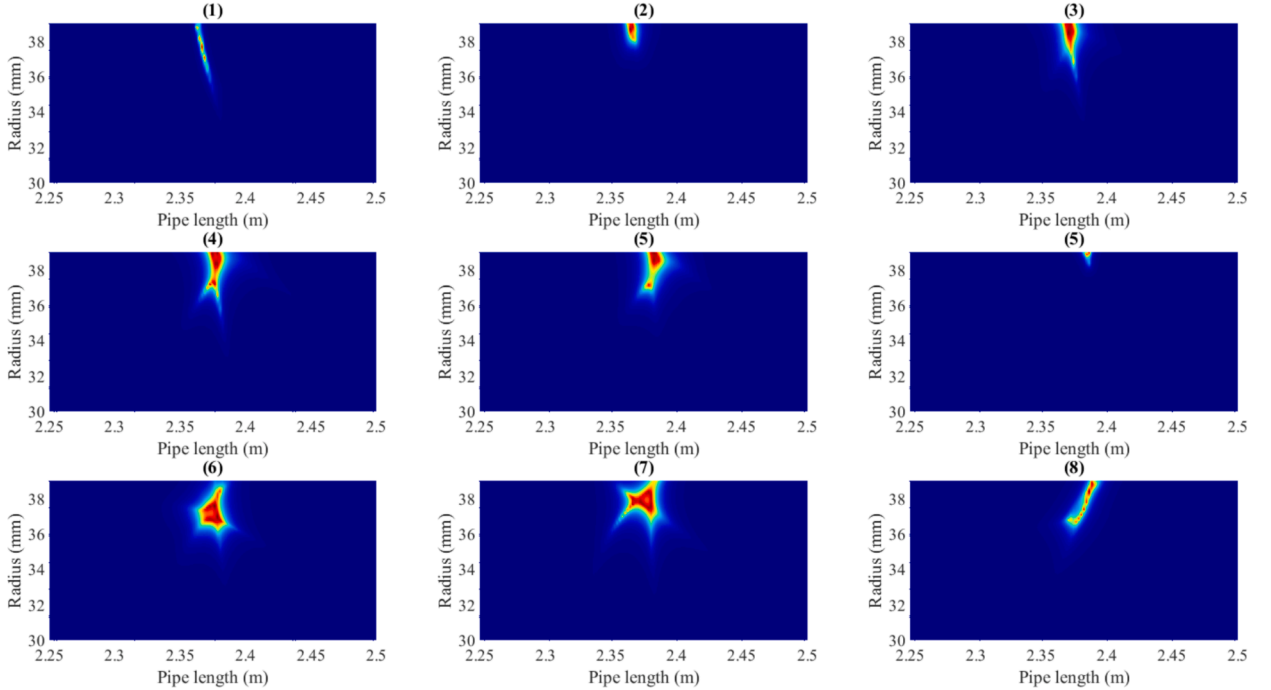


Fig. 5. Local TR-MUSIC images when a Hanning window with width equal to $T/30 = (0.01/30)$ s is applied to the original signal.

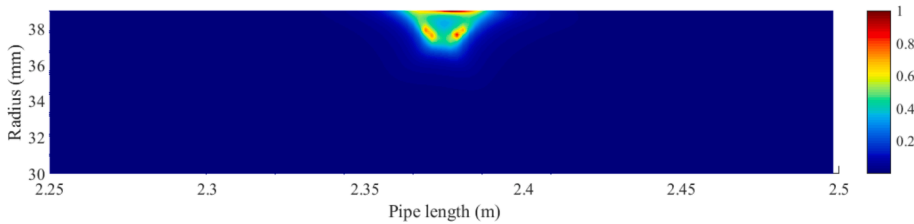


Fig. 6. Combination of the Local TR-MUSIC images.

the blockage locations. In addition, the blockage image as estimated from the time reversal MUSIC has an error in the location and shape of the blockage. The fact that we considered only the most restrictive case of single snapshot data explains why large errors occur (Fig. 1(b)) at relatively low noise levels. To address the noise issue, we conduct the blockage imaging with multiple realizations of noisy data and find that after 20 realizations the output time reversal image remain unchanged. The result of 20 realizations is plotted in Fig. 1 and shows more accurate image. Clearly, the performance can be better, in practice, if one records more than a single realization sample because different noise contributions will statistically cancel out. (See Fig. 8).

To quantitatively estimate the effect of the noise on the proposed pipe imaging methods, we consider the pipe image in the noise free case (Fig. 6) as reference image since it produces accurately the actual blockage. Then for given SNR the imaging error induced by

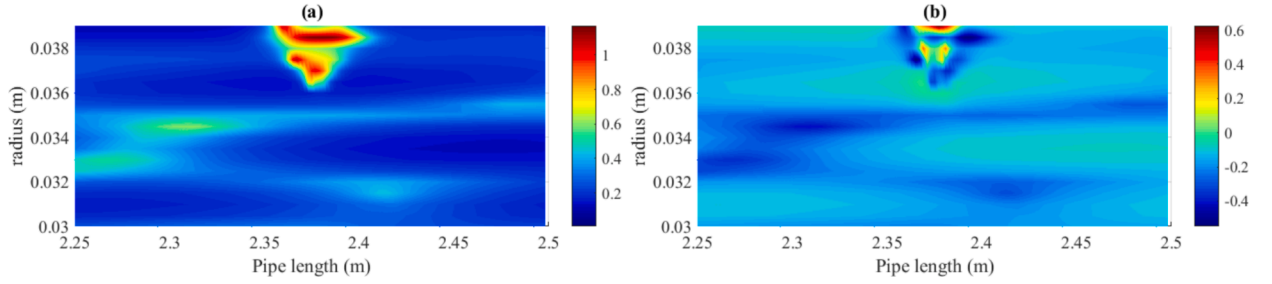


Fig. 7. (a): Plots of the time reversal MUSIC blockage image computed from a single realization noisy synthetic data yielding a signal-to-noise ratio of about 22 dB. (b) Difference between signals for noise-free case.

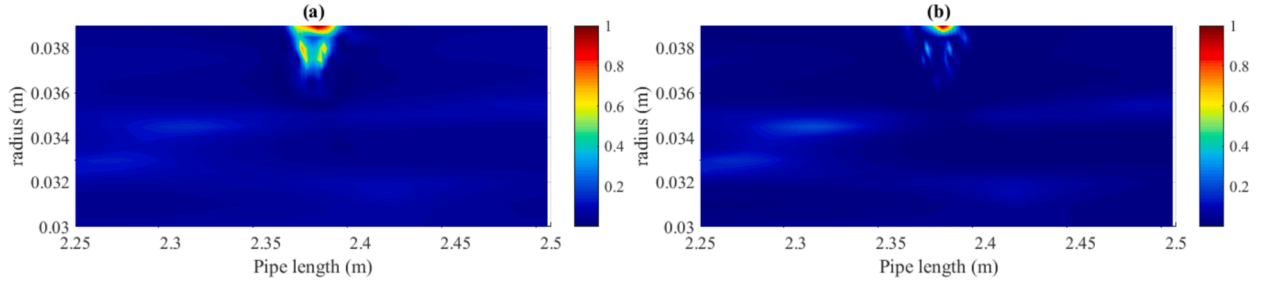


Fig. 8. (a): Plots of the time reversal MUSIC blockage image computed from a 20 realizations noisy synthetic data yielding a signal-to-noise ratio of about 22 dB. (b) Difference between signals for noise-free case.

the noise is evaluated with respect to the reference image. Here, the relative error $e_{x,r}$ is defined by

$$e_{x,r} = \frac{1}{2} \left(\frac{\|\overline{\varphi}_x(\text{SNR}) - \varphi_x(\text{SNR} = 0)\|}{\|\varphi_x(\text{SNR} = 0)\|} + \frac{\|\overline{\varphi}_r(\text{SNR}) - \varphi_r(\text{SNR} = 0)\|}{\|\varphi_r(\text{SNR} = 0)\|} \right) \times 100 \quad (8)$$

where φ_x and φ_r are the projections of the imaging function φ along x and r axis, respectively, and the overline stands for the average. For each SNR value, 20 noise imaging realizations are generated and the relative error between the actual and the ensemble averaged estimated projections of the imaging function φ along x and r axis are computed. Fig. 9 depicts the errors for different SNR considering the blockage explored in the previous section. Fig. 9 clearly shows that the accuracy of the blockage image increases as the SNR increases. Therefore, in practice, such experiment requires accurate equipment and low-noise amplifier to achieve accurate results. This will be further discussed in the experimental section below.

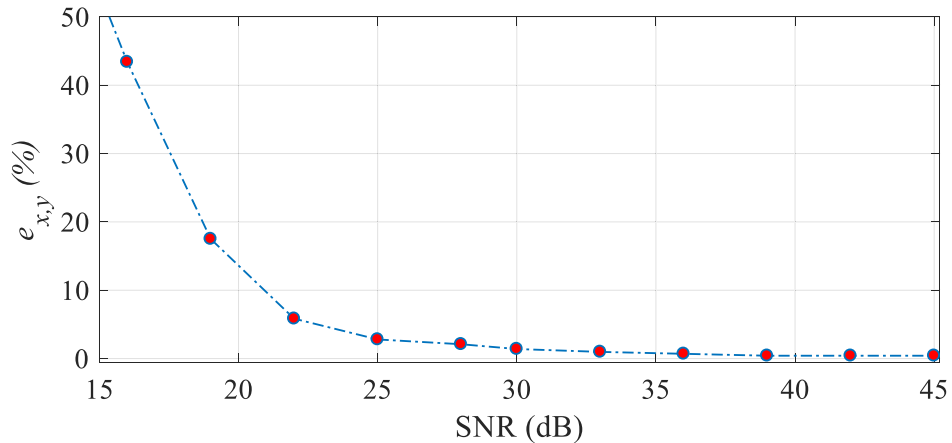


Fig. 9. Imaging error of the blockage sketched in Fig. 3 with respect to SNR (16, 19, 22, 25, 28, 30, 33, 36, 39, 42, 45 dB). The central frequency of the Gaussian source is $f_c = 55$ kHz.

3.3. Synthetic data for elastic pipe case

The proposed method has also been applied for elastic pipe wall test case, and the same geometric parameters of the above numerical example are considered for $\text{SNR} = 30$ dB. The analytical model ([14]) is used to generate synthetic data to test and demonstrate the imaging algorithm in elastic pipe. Three test defect cases are considered, namely (i) internal wall thinning; (ii) stiffness reduction in a small region along the wall; and (iii) external wall thinning. In each case, the defect is located 20 m away from the sensors array. The defect sizes and shapes are shown in Fig. 10a, c, and e. In all test cases, the pipe is considered unbounded, 7.86 cm in diameter, 5.7 mm in thickness, and viscoelastic in material (same as HDPE parameters).

The MIMO data is obtained via an array of 10 transducers placed along the pipe centerline and spaced by 2 cm. The excited wave source signal $S(t)$ is a Gaussian-modulated sine wave with central frequency $f_0 = 70$ kHz (or $\omega_0 = 2\pi f_0$) and a narrow frequency bandwidth $\text{FBW} = [0.9f_0 \text{ to } 1.1f_0]$ [29,30]. The sampling rate is 1 MHz, and the length of the time signal is set to $T = 0.02$ s. The Hanning window width is taken as $T/20$. Applying the proposed imaging algorithm described in the previous section, the pipe images of the three considered test cases are given in Fig. 10b, Fig. 10d, and Fig. 10f. Fig. 10 compares the identified TR image of the defect location, size, and shape, as well as the pipe wall and its thickness, with the actual cases and shows accurately predicted TR images.

4. Testing the proposed TR imaging method using experimental data

4.1. Application in lab facility

The proposed technique for pipeline imaging is experimentally tested in the Water Resources Research Laboratory at Hong Kong University of Science and Technology (HKUST). The experimental setup consists of a straight 6.5 m long, 90 mm diameter HDPE pipe that connects two water-filled tanks at its ends (see Fig. 11). The nominal external diameter of the pipe is equal to 90×10^{-3} m, and the pipe wall thickness is 5.7×10^{-3} m. To emphasize the imaging capability, the experimental setup includes a blockage introduced as a short pipe section located 3.5 m away from the upstream boundary, with a diameter of 33 mm, a thickness $r_B = 7.65$ mm, and a length $l_B = 97$ mm as shown in Fig. 11.

An array of 10 transducer locations is considered at the pipe centerline and spaced by 2 cm, and located 1.5 m away from the upstream boundary. Considering a linear wave system, only two Brüel & Kjær Type 8104 transducers are used; one as a projector and the other as a sensor. The piezoelectric transducers (projector and sensor) are mounted inside the pipeline, with each transducer consisting of a piezoelectric ceramic element housed in a protective, waterproof casing. The projector is stimulated by applying an electrical signal generated by a function generator, typically a gaussian waveform, with a precise central frequency. This voltage signal is sent to the piezoelectric transducers, which convert the electrical energy into mechanical energy, causing the piezoelectric material

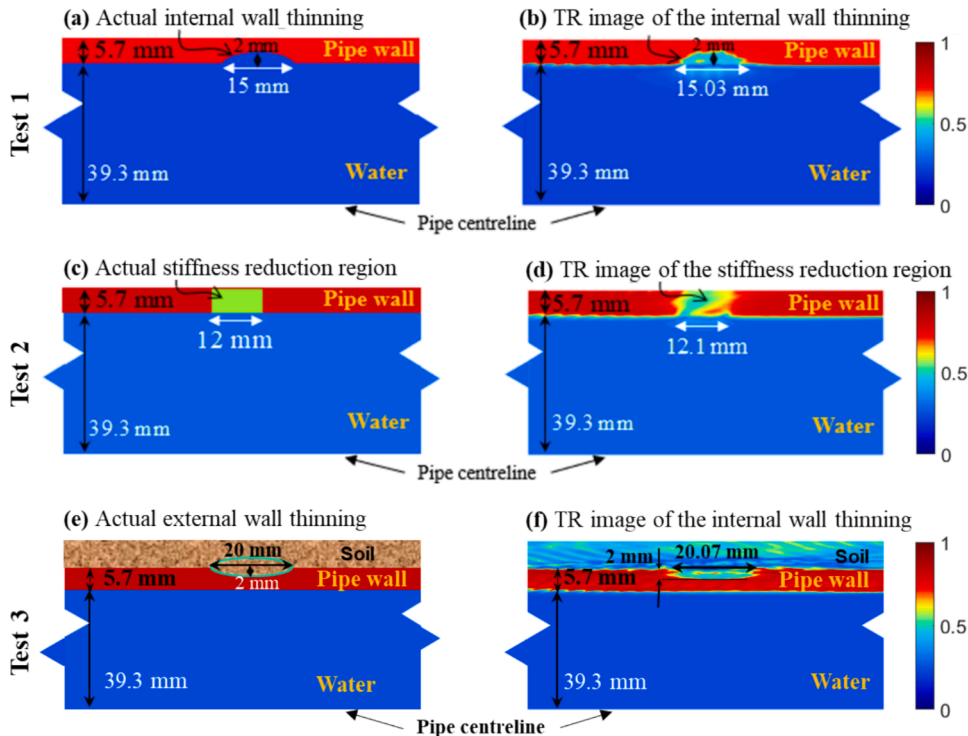


Fig. 10. TR image (a-b) internal pipe wall thinning, (c-d) 10 % reduction in stiffness, and (e-f). The probing central frequency $f_0 = 70$ kHz.

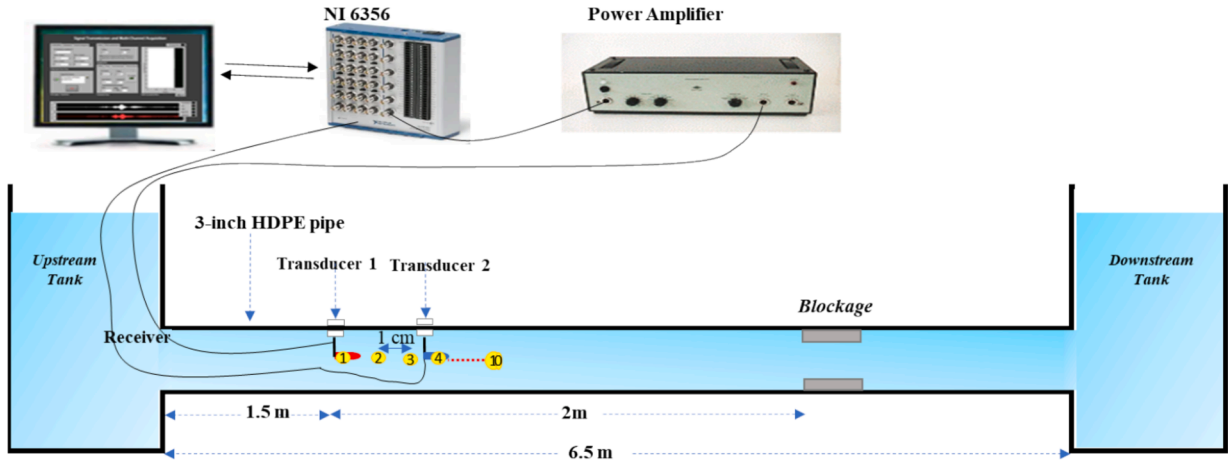


Fig. 11. Sketch map of the high-frequency experimental setup in the Water Resources Research Laboratory at HKUST. 3D printed magnetic base is used to mount the transducers inside a 90 mm HDPE pipe with a total length of 6.5 m. A magnet was mounted to drag the transducers. A blockage is deliberately introduced along the pipe. Experiment facilities consisted of a B&K 2713 power amplifier and a NI 6356 data acquisition board.

to vibrate and generate guided waves that propagate along the pipeline. In this configuration, similar piezoelectric transducer (sensor) is also used to detect the reflected waves. The mechanical vibrations from the reflected waves cause the piezoelectric element to deform, generating an electrical signal that is then captured by the sensor. The received signal is then digitized and processed for further analysis. The multi-input, multi-output (MIMO) data is obtained by successively fixing the projector at a given position and measuring the system response moving the sensor successively through the locations within the array. A Gaussian modulated sine wave source $S(t)$ with central frequency $f_0 = 55$ kHz and a narrow frequency bandwidth $FBW = [0.9f_0 \text{ to } 1.1f_0]$ is used. The sampling rate is 1 MHz, and the length of the measurement time is set to $T = 0.012$ s to prevent the effect of pipe-tank reflection. Prior to data processing, the baseline system response is subtracted to enhance the scattering response signal. The used Hanning window width is one-twentieth the length of the measurement time. Applying the proposed imaging algorithm described in the previous section, which incorporates the Green's function presented in Eq. (3) and evaluated using the same parameters as the experimental setup, we obtain the pipe image shown in Fig. 12. This image clearly reveals the location and size of the blockage, as well as the pipe wall and its thickness, demonstrating the algorithm's accuracy.

4.2. Application in field-scale facility (Beacon Hill)

The imaging algorithm is further validated in a field-scale facility which represents a more realistic system scale. The facility comprises a 250 m long water-filled viscoelastic 160 mm HDPE pipe (with radius $R = 73$ mm) (see Fig. 13). A 36.5 m long straight section indicated in Fig. 13 is selected to perform the imaging test. The section ends connect to two T-access points from which transducers can be introduced. In this case, the transducers are introduced from the downstream end T-access point (T1). Along this section, three T-connections exist (T 2, T3, and T4), and a blockage with a thickness $r_B = 16$ mm, and a length $l_B = 95$ mm, is introduced at 7 m away from the location $x = 0$ m (see Fig. 13).

Applying the imaging algorithm, which incorporates the Green's function presented in Eq.3 and evaluated using the same parameters of the pipe system illustrated in Fig. 13, and following the measurement procedure conducted in the previously described lab case, the pipe section image is provided in Fig. 14. The image indicates the pipe wall and its thickness, along which three pipe wall openings are shown corresponding to the existing three T-connections. Moreover, the blockage is accurately identified in both location and size.

5. Conclusion

This paper proposed a computationally efficient, nondestructive, and medium range guided wave imaging of pressurized water supply lines based on the time reversal (TR) technique. The main conclusion points are:

- The proposed technique is non-disruptive and non-intrusive. It requires a send-receive transducer to be inserted in the water column from a single access point (eg. an inspection-T) and used to measure multi-input-multi-output (MIMO) pressure wave signals that propagate along the pipeline.
- High resolution imaging is achieved using TR-MUSIC approach in the frequency range of 10 kHz to 100 kHz.
- The proposed imaging technique applied with synthetic data is able to reveal the pipe wall inner and outer condition in millimeter scale.

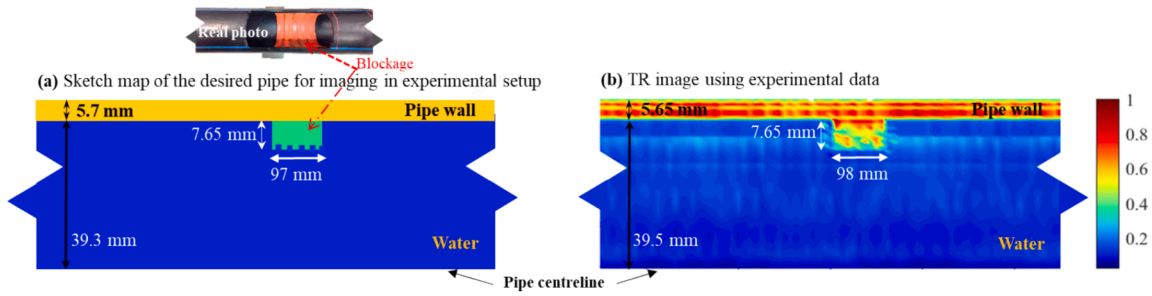


Fig. 12. Imaging results of the considered pipeline system. The image is created using the experimental data from the configuration given in Fig. 11. The image is computed at $f_0 = 55$ kHz, and the signal-to-noise ratio $\text{SNR} = 35$ dB.

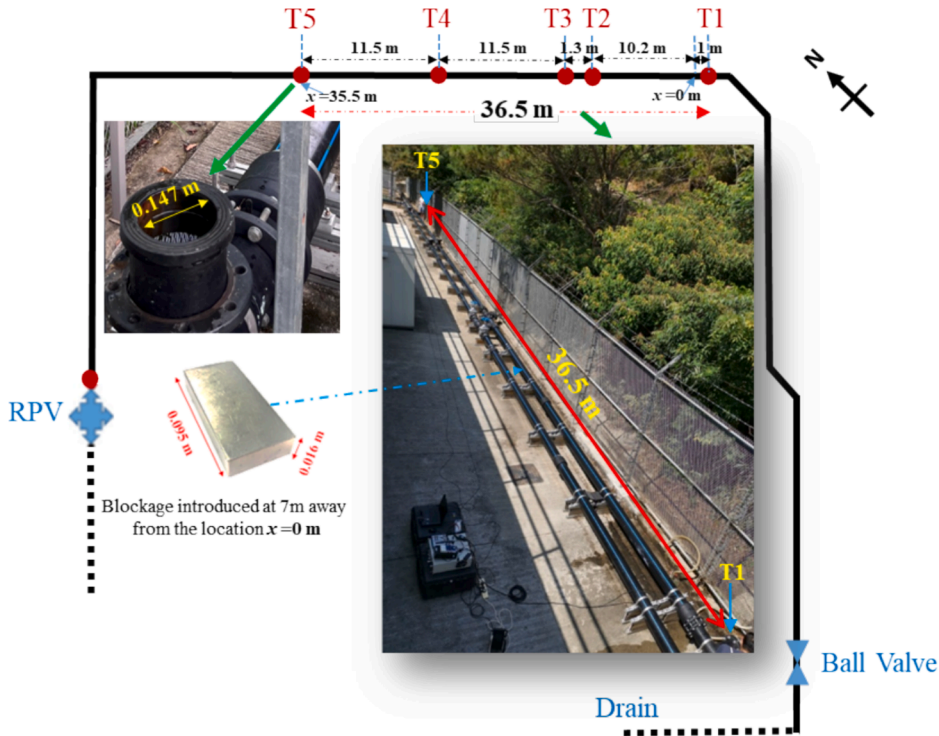


Fig. 13. Setup of the pipe transient experiment in Beacon Hill, Hong Kong.

- Both lab and field-scale facilities are used to validate the proposed imaging technique. In the lab, we successfully imaged a straight 6.5 m long, water-filled 90 mm HDPE pipe containing a small blockage with thickness 7.65 mm and a length 97 mm. In the field-scale facility, we accurately imaged a 36.5 m long section water-filled 160 mm HDPE pipe containing three T-connections and a small blockage with thickness 16 mm and length 95 mm.

The development of TR-imaging for water-filled pipelines would permit UWSS management to move from reactive to proactive mode, where anomalies with length scales of the order of millimetre are detected and imaged (not just localised). Thus, it serves as a decision support tool for timely maintenance and mitigation measures.

CRediT authorship contribution statement

Saber Nasraoui: Writing – original draft, Methodology, Data curation, Conceptualization. **Moez Louati:** Validation. **Mohamed Salah Ghidaoui:** Supervision, Funding acquisition.

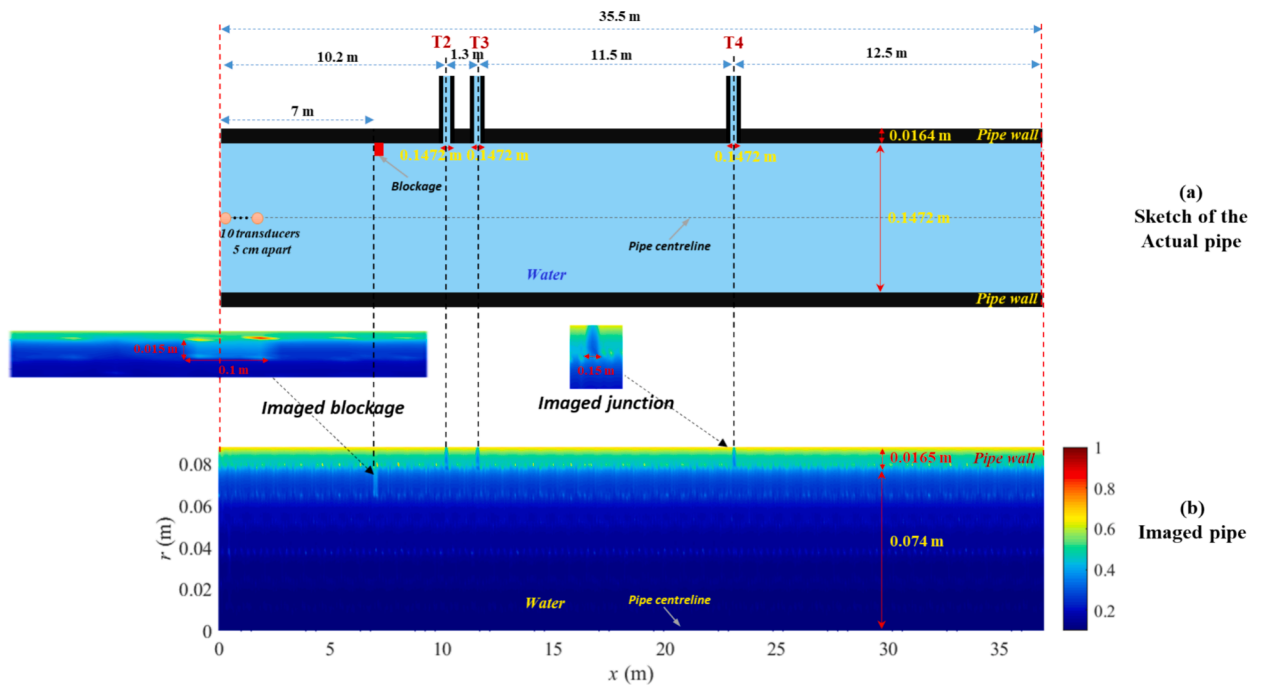


Fig. 14. Imaging results of the considered pipeline in Beacon Hill, Hong Kong. The image is computed at $f_0 = 75$ kHz, and the signal-to-noise ratio $\text{SNR} = 30$ dB.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] R. Liemberger, A. Wyatt, Quantifying the global non-revenue water problem, *Water Supply* 19 (3) (2018) 831–837.
- [2] B. Nestleroth, S. Flamberg, W. Condit, J. Matthews Battelle, L. Wang, A. Chen, Field Demonstration of Innovative Condition Assessment Technologies for Water Mains: Leak Detection and Location. National Risk Management Research Laboratory - Office of Research and Development - U.S. Environmental Protection Agency, Cincinnati, Ohio, 2012.
- [3] “SmartBall PWA Inspection Report 16-inch Hanover Park Force Main, prepared by Pure Technologies U.S., Inc., dated November 2015”.
- [4] “SmartBall Inspection Report NTW 700mm Pipeline Prepared for Water Supplies Department (Hong Kong) Prepared by Pure Technologies October, 2018”.
- [5] X. Wang, M.S. Ghidaoui, Pipeline leak detection using the matched-field processing method, *ASCE, J. Hydraul. Eng.* 144 (6) (2018) 04018030.
- [6] M. Waqar, M. Louati, M.S. Ghidaoui, Time-reversal of water hammer waves, *J. Hydraul. Res.* 1 (60) (2022) 25–45.
- [7] F. Zouari, M. Louati, E. Blåsten and M. S. Ghidaoui, “Multiple defects detection and characterization in pipes,” *BHR Group 2018 Pressure Surges* 13.
- [8] P. Kumar, P.K. Mohapatra, Partial blockage detection in pipelines by modified reconstructive method of characteristics technique, *J. Hydraul. Eng.* 148 (4) (2022) 04022003.
- [9] A.M. Sattar, H.M. Chaudhry, Leak detection in pipelines by frequency response method, *J. Hydraul. Res.* 46 (sup1) (2008) 138–151.
- [10] S. Kim, Multiple discrete blockage detection function for single pipelines, *Proceedings* 2 (11) (2018) 582.
- [11] S. Meniconi, B. Brunone, M. Ferrante, C. Capponi, C.A. Carrettini, C. Chiesa, D. Segalini, E.A. Lanfranchi, Anomaly pre-localization in distribution–transmission mains by pump trip: preliminary field tests in the Milan pipe system, *J. Hydroinf.* 17 (3) (2015) 377–389.
- [12] T.C. Che, H.F. Duan, P.J. Lee, Transient wave-based methods for anomaly detection in fluid pipes, *Mech. Syst. Sig. Process.* 160 (2021) 107874.
- [13] F. Zouari, E. Blåsten, M. Louati, M.S. Ghidaoui, Internal pipe area reconstruction as a tool for blockage detection, *J. Hydraul. Eng.* 154 (6) (2019) 04019019.
- [14] S. Nasraoui, M. Louati, M.S. Ghidaoui, Blockage detection in pressurized water-filled pipe using high frequency acoustic waves, *Mech. Syst. Signal Proc.* 185C (2022) 109817.
- [15] S. Nasraoui, M. Louati, M.S. Ghidaoui, High resolution acoustic identification of clusters of small blockages in fluid-filled pipe using maximum likelihood estimation, *J. Acoust. Soc. Am.* 153 (5) (2023) 3086.
- [16] G. Grigoropoulos, M.S. Ghidaoui, M. Louati, S. Nasraoui, Time reversal of waves in hydraulics: experimental and theoretical proof with applications, *J. Hydraul. Res.* 1 (2022) 60.
- [17] A. Dubey, Z. Li, P.J.A.R. Murch, Measurement and characterization of acoustic noise in water pipeline channels, *IEEE Access* 7 (2019) 56890–56903.
- [18] H. Lev-Ari and A. J. Devaney, *IEEE Sensor Array and Multichannel Signal Processing Workshop, Cambridge (MA), USA*, p. 509–513, 2000.
- [19] C. Prada, S. Manneville, D. Spoliansky, M. Fink, Decomposition of the time reversal operator: detection and selective focusing on two scatterers, *J. Acoust. Soc. Am.* 99 (1996) 2067–2076.
- [20] A. Baussard, T. Boutin, Time-reversal RAP-MUSIC imaging, *Waves Random Complex Media* 18 (1) (2008) 151–160.

- [21] C. Fan, M. Pan, F. Luo, B.W. Drinkwater, Multi-frequency time-reversal-based imaging for ultrasonic nondestructive evaluation using full matrix capture, *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* 61 (12) (2014) 2067–2074.
- [22] J.J. Sakurai, *Modern Quantum Mechanics*, Addison, New York, 1994.
- [23] Y. Jie, C.L. Anne, G. Bernhard, H. Fazle, Inverse scattering theory: Inverse scattering series method for one dimensional non-compact support potential, *J. Math. Phys* 55 (12) (2014) 123512.
- [24] S. Rienstra, A. Hirschberg, *An introduction to Acoustics*, Self-publishing, Eindhoven, 2019.
- [25] S. Rienstra, B. Teste, An analytic Greens' function for a lined circular duct containing uniform mean flow, *J. Sound Vib.* 317 (5) (2008) 994–1016.
- [26] T.C. Lin, G.W. Morgan, Wave propagation through fluid contained in a cylindrical, elastic shell, *J. Acoust. Soc. Am.* 28 (6) (1956) 1165–1176.
- [27] A.J. Devaney, E.A.M.a.F.K. Gruber, Time-reversal-based imaging and inverse scattering of multiply scattering point targets, *J. Acoust. Soc. Am.* 118 (2005) 3129.
- [28] S. Fan, A. Zhang, H. Sun, F. Yun, A local TR-MUSIC algorithm for damage imaging of aircraft structures, *Sensor* 21 (10) (2021) 3334.
- [29] M. Louati, M.S. Ghidaoui, High frequency acoustic wave properties in a water-filled pipe. part 1: Dispersion and multi-path behaviour, *J. Hydraul. Res.* 55 (5) (2017) 613–631.
- [30] Z. Lai, M. Louati, S. Nasraoui, M.S. Ghidaoui, Numerical investigation of high frequency wave-leak interaction in water-filled pipes, *J. Hydraul. Eng* 147 (1) (2021) 04020091.